

5.3.1 Polar Worksheet

1. Consider the polar curve $r = 2 - \sin(\theta)$.

a) Parameterize the polar curve.

b) At what value of θ between $[0, 2\pi]$ does the curve reach the point $(x, y) = (0, 1)$?

2. Consider the polar equation $r = 2 \sin(3\theta)$.

a) Find a formula for $\frac{dy}{dx}$. No simplification necessary.

b) Determine the value of θ between $\left(\frac{\pi}{2}, \pi\right)$ when the curve reaches the point $(0, 0)$?

c) Determine the slope of the tangent line at the value of θ found in part (b).

d) Determine the *equation* of the tangent line when $\theta = \frac{\pi}{4}$.

3. Find the points on the curve given by $r = 3\sin(\theta)$ from $0 \leq \theta \leq 2\pi$ where there are horizontal & vertical tangent lines and singular points.

4. Find the points on the curve given by $r = 1 - \sin(\theta)$ from $0 \leq \theta \leq 2\pi$ where there are horizontal & vertical tangent lines and singular points.

5. Find the points on the curve given by $r = e^\theta$ from $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$ where there are horizontal & vertical tangent lines and singular points.

6. Find the points on the curve given by $r = 2\cos^2(\theta)$ from $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$ where there are horizontal & vertical tangent lines and singular points.

7. Determine the exact length of each of the following polar curves.

a) $r = 2 \cos(\theta), 0 \leq \theta \leq \pi$.

b) $r = 5^\theta, 0 \leq \theta \leq 2\pi$

c) $r = \theta^2, 0 \leq \theta \leq 2\pi$

d) A single tracing of $r = 2[1 + \cos(\theta)]$

e) A single tracing of $r = 2\sin(3\theta)$. Do not evaluate this integral.